Bharadwaj Ramachandran

Applied AI and Enterprise Engineering | Senior Manager, AI Solutions

www.bharadwajramachandran.com | LinkedIn | GitHub

Senior Manager, AI Solutions at Thermo Fisher Scientific, with 12+ years across software engineering, enterprise automation, and data engineering. Leads eight engineers delivering AI across business groups, combining team development, technical direction, and executive communication.

Thermo Fisher Scientific

Senior Manager, AI Solutions | Dec 2025 - Present

Promoted December 7, 2025.

- Lead technical delivery across business groups for projects taken on within one quarter, with combined expected annual revenue of $10-15M.
- Scaled the engineering team from two to eight within one year to deliver enterprise AI solutions.
- Set technical direction for the team’s work on model routing, reusable agent architectures, and an AI gateway.
- Led rollout of a reusable Model Context Protocol (MCP) architecture and standards for development, deployment, and integration with internal and external clients; services supported 10,000 users and recorded 200,000+ tool calls in 2026 to date.
- Directed development of a custom context layer that achieved 90%+ accuracy in applying context to SQL query generation for data retrieval; broader rollout planned.

Staff Software Engineer | Nov 2023 - Dec 2025

- Led development of a field-service application combining instrument manuals and service history to recommend next actions, saving more than $1M annually in service visits.
- Built a multi-agent system to guide manufacturing decisions, supporting a 43% reduction in excess and obsolete (E&O) levels.
- Replaced Power BI reporting with multi-agent querying and analysis, reducing costs by 30% through lower Power BI licensing spend.
- Built a GenAI playground to compare model quality, execution time, and cost; developed stateful assistant orchestration using Jupyter notebooks.
- Worked closely with OpenAI teams on platform integration into Thermo Fisher workflows; presented AI solutions to the CFO, CTO, CIO, and other executives.

Globality

Senior Software Engineer | May 2021 - May 2023

- Managed the team’s product roadmap and delivery commitments; partnered with engineering and business stakeholders to identify needs and resolve operational gaps.
- Established technical design reviews across the software development lifecycle, platform changes, and third-party integrations; standardized release documentation and automated quality checks.
- Created support-intake processes and administrative tools to help frontline teams investigate root causes and resolve client-reported issues.
- Evaluated build-versus-buy options against cost, scalability, and business needs; implemented enterprise integrations and Active Directory-based onboarding.

Professional Experience Continued

Kaiser Permanente

Software Engineer | Oct 2018 - Sep 2021

- Delivered Angular applications, services, and Adobe Experience Manager components in partnership with product managers, designers, and business stakeholders.
- Reviewed code and end-to-end functionality, contributed to service architecture, and promoted test automation throughout the development lifecycle.
- Mentored developers and trained teams to maintain SSIS integrations; created a departmental training and internal certification program.

Infostretch

Automation Engineer | Oct 2015 - Sep 2018

- Built a reusable Kaiser Permanente testing framework to standardize automation across teams; automated approximately 94% of test cases.
- Integrated automated tests into deployment workflows and built scripts to publish test results in Quality Center, giving engineering and business teams visibility into progress and failures.

UST Global

Senior Systems Analyst (full time) | Oct 2014 - Sep 2015

Data Engineer (contract) · Jul 2013 - Oct 2014

- Built a data-maintenance interface that reduced business change requests and saved nearly $300,000 annually.
- Developed SQL and SSIS pipelines for vendor-file ingestion, warehouse-to-mart data movement, and analytics; trained offshore engineers and organized data-change delivery.

Technical Skills and Leadership

Engineering leadership: Team growth & development; Cross-functional delivery; Technical direction & design reviews; Roadmap & delivery management; Stakeholder & executive communication; Support intake & operational processes

Applied AI: Multi-agent systems, reusable agent architectures, Microsoft AutoGen, Jupyter notebooks; Model Context Protocol development, deployment, and client integration; Model comparison across quality, execution time, and cost; model routing; RAG, Azure AI Search, and context for SQL query generation; Technical direction for the team’s AI gateway; OpenAI APIs and agents

Software engineering: REST APIs and microservices; React, Angular, Adobe Experience Manager; Python, JavaScript, TypeScript, Go; Code reviews, automated quality checks, and test automation

Data & integration: PostgreSQL, Microsoft SQL Server, MySQL; SQL, PL/SQL, Redis; SQL Server Integration Services (SSIS), MuleSoft

Platforms & delivery: Kubernetes; Jenkins, CircleCI, Harness, Drone; Git

Identity & access: Microsoft Entra ID (formerly Azure AD)

Observability: Grafana

Open Source and Technical Projects

- Contributed an assistant-agent example connecting function calling to Azure AI Search for RAG, with Azure Identity-based index-access checks. Merged PR #2594
- Developed an example implementation using Model Hardware Standard (MHS).

Education and Professional Development

Professional Certificate in Machine Learning and Artificial Intelligence
UC Berkeley Executive Education | Completed March 2026 | Credential

Bachelor’s degree in Computer Science
Manav Bharti University | 2008-2012